FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ

PHYSICS

ELECTROSTATIC POTENTIAL AND CAPACITANCE

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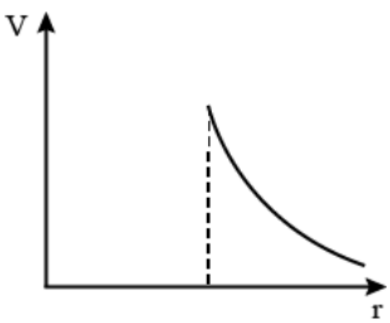
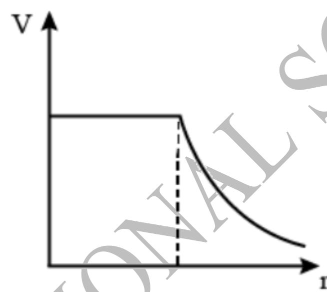
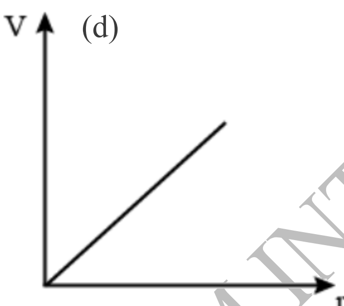
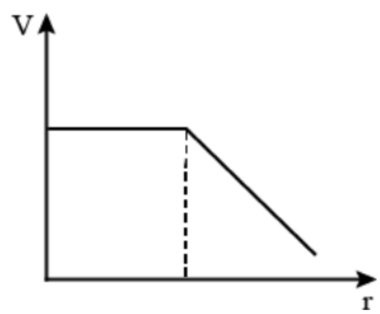
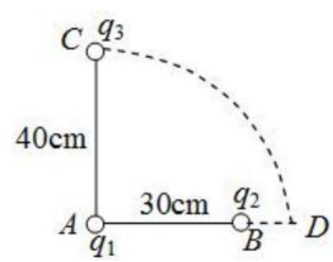
1. The electric potential at the surface of an atomic nucleus (Z=50) of radius 9.0 x 10^{-13} cm is
   (a) 9 x 10^5 V (b) 8 x 10^6 V (c) 80 V (d) 9 V
2. In the case of a charged metallic sphere, potential (V) changes with respect to distance (r) from the centre as
   (a)
   

   (b)
   

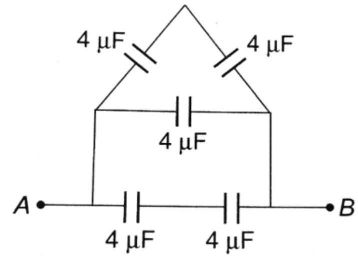
   (c)
   

   (d)
   

3. In bringing an electron towards another electron, the electrostatic potential energy of the system
   (a) increases (b) decreases (c) remains unchanged (d) becomes zero
4. The potential on the hollow sphere of radius 1 m is 1000 V, then potential at ¼ m from the centre of the sphere is
   (a) 1000 V (b) 500 V (c) 250 V (d) zero V
5. Two charges q_1 and q_2 are placed 30 cm apart as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of the system is $\frac{q_3 k}{4\pi\varepsilon_0}$, where k is
   

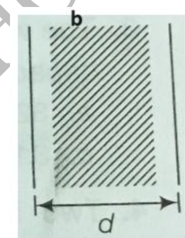
   (a) 8 q_2 (b) 8 q_1 (c) 6 q_2 (d) 6 q_1

6. Equivalent capacitance between A and B is



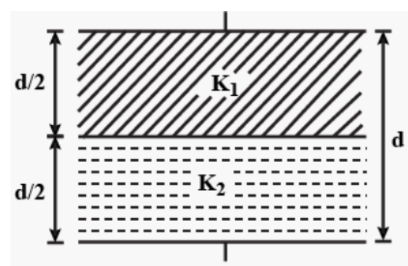
(a) 8 μF (b) 6 μF (c) 268 μF (d) 10/38 μF

7. A copper plate of thickness b is placed inside a parallel capacitor of plate distance ‘d’ and area ‘A’ as shown in the figure. The capacitance of the capacitor is



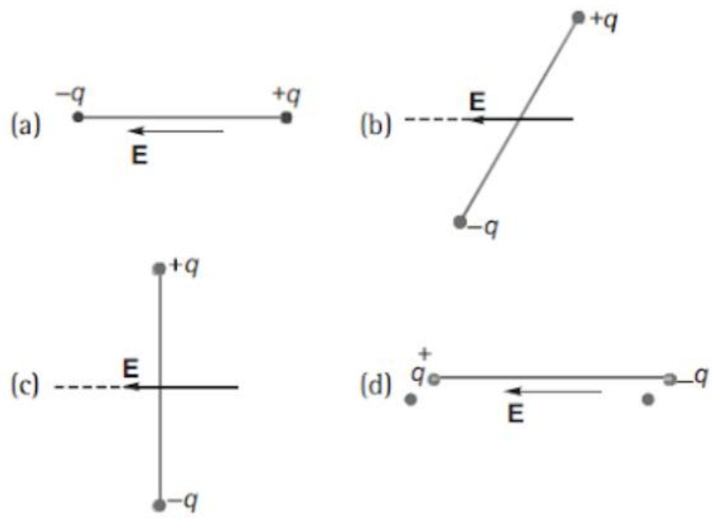
(a) $\frac{A\varepsilon_0}{d}$ (b) $\frac{A\varepsilon_0}{b}$ (c) $\frac{A\varepsilon_0}{d-b}$ (d) infinity

8. A parallel plate capacitor of plate area A and separation d is filled with dielectric as shown in the figure. The dielectric constants are K1 and K2. Net capacitance is

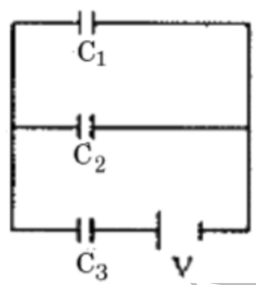


(a) $\frac{\varepsilon_0 A}{d}(k_1+k_2)$ (b) $\frac{\varepsilon_0 A}{d}\left(\frac{k_1+k_2}{k_1k_2}\right)$ (c) $\frac{2\varepsilon_0 A}{d}\left(\frac{k_1k_2}{k_1+k_2}\right)$ (d) $\frac{2\varepsilon_0 A}{d}\left(\frac{k_1+k_2}{k_1k_2}\right)$

9. In which of the states shown in the figure, is potential energy of an electric dipole maximum?



10. Find the voltage across C_3.



(a) $\frac{(c_1+c_2)V}{c_1+c_2+c_3}$ (b) $\frac{c_1V}{c_1+c_2+c_3}$ (c) $\frac{c_2V}{c_1+c_2+c_3}$ (d) $\frac{c_3V}{c_1+c_2+c_3}$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- **A. Both Assertion and Reason are true and Reason is correct explanation of Assertion.**
- **B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.**
- **C. Assertion is true but Reason is false.**
- **D. Both Assertion and Reason are false.**

11. **Assertion:** Increasing the charge on the plates of a capacitor means increasing the capacitance.
Reason: Capacitance is directly proportional to charge.
12. **Assertion:** Dielectric polarization means formation of positive and negative charges inside the dielectric.
Reason: Free electrons are formed in this process.
13. **Assertion:** When air between the plates of a parallel plate condenser is replaced by an insulating medium of dielectric constant its capacity increases.
Reason: Electric field intensity between the plates with dielectric in between it is reduced.
14. **Assertion:** The electrostatic force between the plates of a charged isolated capacitor decreases when dielectric fills whole space between plates.

Reason: The electric field between the plates of a charged isolated capacitance increases when dielectric fills whole space between plates.

15. **Assertion:** In the absence of an external electric field, the dipole moment per unit volume of a polar dielectric is zero.

    Reason: The dipoles of a polar dielectric are randomly oriented.